HAND & UPPER LIMB · NEET-SS / FET / INI-SS

Carpal instability and scaphoid

Part A: Scaphoid fractures

1. Anatomy

- **Scaphoid:** the **link between the proximal and distal carpal rows**; **about 80% covered by articular cartilage**; regions: **proximal pole, waist, distal pole, tubercle**.
- **Blood supply (Gelberman): dorsal carpal branch of the radial artery** enters the **dorsal ridge (non-articular) at the waist**, supplying **about 70-80% (including the proximal pole) by retrograde flow**; **superficial palmar branch** supplies the **distal 20-30% (tubercle)**. **Retrograde supply: proximal pole fractures have the highest AVN/non-union risk.**
- **Commonest carpal bone fractured (about 60-70% of carpal fractures)**; **young men**; **FOOSH (wrist dorsiflexed over about 95 degrees and radially deviated)**.

2. Diagnosis

- **Anatomical snuffbox tenderness** (sensitive, not specific), **scaphoid tubercle tenderness**, **pain on axial compression of the thumb**; **combination of signs** improves specificity.
- **Radiographs: scaphoid series (PA, lateral, PA with ulnar deviation (scaphoid view), semi-pronated oblique)**: **up to about 20% of fractures are occult initially**.
- **Clinical suspicion with normal radiographs: immobilise and image**: **MRI (gold standard for occult fractures: sensitivity and specificity near 100%; best within the first days)**, **CT** (displacement, union, humpback), **bone scan** (72 hours; sensitive,

not specific). **Early MRI is cost-effective** (avoids unnecessary immobilisation).

- **CT along the long axis of the scaphoid**: assess **displacement** (**over 1 mm**), **humpback deformity** (**intrascaphoid angle over 35 degrees on sagittal CT**; **height-to-length ratio over 0.65**), comminution.
- **Displacement criteria (unstable): over 1 mm step or gap**, **intrascaphoid angle over 35 degrees**, **radiolunate angle over 15 degrees (DISI)**, **comminution**, **associated perilunate injury**, **proximal pole fracture**.

3. Classification

- **Herbert classification:**
  - **A (stable acute): A1 tubercle**, **A2 incomplete waist**,
  - **B (unstable acute): B1 distal oblique**, **B2 complete waist**, **B3 proximal pole**, **B4 trans-scaphoid perilunate fracture-dislocation**, (B5 comminuted),
  - **C: delayed union**, **D: non-union** (D1 fibrous, D2 sclerotic/pseudarthrosis).
- **Russe classification (orientation): horizontal oblique, transverse, vertical oblique (least stable: shear)**.
- **Location: waist (about 65-70%)**, **proximal pole (about 10-20%)**, **distal pole/tubercle (about 10-15%: heal well)**.

4. Management

- **Tubercle and incomplete fractures: short below-elbow cast or splint 4-6 weeks**.
- **Undisplaced waist fractures: cast immobilisation (below-elbow; thumb inclusion not necessary)** for **6-8 weeks** (**union about 90%+**) vs **percutaneous screw fixation** (**faster return to work/sport**, **SWIFFT trial (UK, 2020): no difference in function at 1 year**

between surgery and cast with early fixation of non-unions; **cast with CT at 6 weeks and fixation of non-unions is the standard**).

- **Displaced waist fractures (over 1-2 mm): ORIF/percutaneous fixation** (**non-union about 50% with cast**).
- **Proximal pole fractures: fixation (dorsal approach)** (**high non-union risk ~30% with casting**).
- **Fixation: headless compression screw (Herbert/Acutrak)** along the **central axis of the scaphoid** (**central placement = strongest, fastest union**); **volar approach** for waist/distal fractures (**preserves the dorsal blood supply**), **dorsal approach** for proximal pole fractures (**risk to the dorsal ridge vessels; wrist flexion to access the entry point**).
- **Paediatric scaphoid fractures:** usually **distal third**, heal well with casting.

5. Scaphoid non-union

- **Non-union about 5-10%** overall; **risk factors: displacement**, **proximal pole**, **delay in diagnosis/treatment (over 4 weeks)**, **smoking**, **inadequate immobilisation**, **vertical oblique pattern**, **associated carpal instability**.
- **Natural history: SNAC wrist (scaphoid non-union advanced collapse)**: predictable progression:
  - **Stage I: radial styloid-scaphoid arthritis** (distal scaphoid fragment and radial styloid),
  - **Stage II: + scaphocapitate arthritis**,
  - **Stage III: + capitolunate (midcarpal) arthritis** (**radiolunate joint spared**: **proximal pole moves with the lunate**),
  - (Stage IV: pancarpal/radiolunate).
- **Humpback deformity**: **palmar flexion of the distal fragment + dorsal extension of the proximal fragment (with the lunate): DISI**.

- **Treatment of non-union (no arthritis):**
  - **Stable, undisplaced, fibrous non-union without humpback**: **percutaneous screw ± bone graft**.
  - **Humpback/displaced: open reduction, volar wedge (corticocancellous) graft from the iliac crest (Fisk-Fernandez), screw fixation**.
  - **Proximal pole AVN (no punctate bleeding at surgery: the best indicator of viability; MRI with gadolinium)**: **vascularised bone graft**: **1,2-intercompartmental supraretinacular artery (1,2-ICSRA) pedicled graft (Zaidemberg, dorsal distal radius)**, **medial femoral condyle free vascularised graft (MFC; for humpback with AVN)**, **volar pronator quadratus pedicle graft (Kuhlmann)**.
  - **Matti-Russe** (volar inlay cancellous graft: historical).
- **SNAC with arthritis (salvage): radial styloidectomy** (stage I), **scaphoid excision + four-corner fusion (lunate, capitate, hamate, triquetrum: stages II-III)**, **proximal row carpectomy (PRC: requires intact capitate head and lunate fossa cartilage: stage I-II; contraindicated if capitolunate arthritis)**, **total wrist arthrodesis** (stage IV / pancarpal), **wrist denervation** (AIN/PIN neurectomy).

Part B: Carpal instability

6. Kinematics and ligaments

- **Proximal row = intercalated segment** (no tendon insertions): moves in response to forces. **Scaphoid tends to flex**, **triquetrum tends to extend** (helical hamate-triquetral articulation); the **lunate is balanced between them** by the **scapholunate (SL)** and **lunotriquetral (LT)** ligaments.

- **Scapholunate interosseous ligament (SLIL): dorsal part strongest (most important)**, proximal (membranous), palmar. **Secondary stabilisers: dorsal intercarpal ligament (DIC)**, **dorsal radiocarpal ligament (DRC)**, **scaphotrapeziotrapezoid (STT) ligaments**, **radioscaphocapitate (RSC)**.
- **Lunotriquetral ligament: palmar part strongest.**
- **Extrinsic palmar ligaments: radioscaphocapitate**, **long radiolunate**, **short radiolunate**, **radioscapholunate (ligament of Testut: neurovascular, not mechanical)**, **ulnolunate, ulnotriquetral**. **Space of Poirier** (**weak area between the RSC and long radiolunate, over the capitolunate joint**: **perilunate dislocations escape through it**).
- **Normal angles (lateral radiograph): scapholunate angle 30-60 degrees (mean 47)**, **radiolunate angle 0 ± 15 degrees**, **capitolunate angle 0-15 degrees**.

7. Classification of carpal instability (Mayo)

- **CID (carpal instability dissociative)**: within a row: **SL dissociation (DISI)**, **LT dissociation (VISI)**, unstable scaphoid fractures.
- **CIND (non-dissociative)**: between rows or radius-carpus: **midcarpal instability**, **radiocarpal instability** (after malunited distal radius fractures, ulnar translocation (rheumatoid)).
- **CIC (complex)**: **perilunate dislocations** (CID + CIND).
- **CIA (adaptive)**: carpal malalignment from **distal radius malunion** (dorsal tilt: adaptive DISI).

8. Scapholunate dissociation (commonest carpal instability)

- **Mechanism: FOOSH with extension, ulnar deviation and intercarpal supination**; **Mayfield stage I** of perilunate injury.
- **Signs:** dorsal SL tenderness (**just distal to Lister's tubercle**), **Watson scaphoid shift test** (**thumb pressure on the scaphoid**

tubercle while moving from ulnar to radial deviation: painful clunk as the scaphoid subluxes dorsally over the rim and reduces).

- **Radiographs:**
  - **SL gap over 3 mm (Terry Thomas / David Letterman sign)** (**clenched-fist PA view** (dynamic) and comparison),
  - **Scaphoid ring (cortical ring) sign** (**flexed, foreshortened scaphoid; ring under 7 mm from the proximal pole**),
  - **SL angle over 60-70 degrees**, **DISI (dorsal intercalated segment instability: lunate extended: radiolunate angle over 15 degrees dorsally)**.
- **Investigations: MR arthrography**, **wrist arthroscopy (gold standard)**: **Geissler classification** (**I: attenuation/haemorrhage, no incongruity; II: midcarpal step-off, probe can enter; III: step-off in both radiocarpal and midcarpal joints, probe passes between bones; IV: gross instability, 2.7 mm arthroscope passes through the gap ("drive-through sign")**).
- **Garcia-Elias staging** (6-question algorithm: partial tear? repairable dorsal SL? scaphoid normally aligned? carpal malalignment reducible? cartilage normal?).
- **Treatment:**
  - **Acute (under about 6 weeks) partial (Geissler I-II): cast, arthroscopic debridement ± thermal shrinkage, percutaneous pinning (SL and SC K-wires)**.
  - **Acute complete with repairable ligament: open repair (suture anchors) + K-wire stabilisation + dorsal capsulodesis (Blatt, Mayo DIC-based)**.
  - **Chronic, reducible, no arthritis: reconstruction**: **3-ligament tenodesis (3LT; Garcia-Elias: modified Brunelli FCR tenodesis)**, **SLAM**, **RASL (reduction and association of the scaphoid and lunate: Herbert screw across the SL joint)**, **capsulodesis**.

- **Chronic irreducible/fixed, no arthritis: STT fusion**, **scaphocapitate fusion** (limited).
- **Arthritis: SLAC wrist**.

9. SLAC wrist (scapholunate advanced collapse) (Watson)

- **Commonest pattern of degenerative wrist arthritis.**
- **Stage I: radial styloid-scaphoid**, **Stage II: entire radioscaphoid fossa**, **Stage III: capitolunate (midcarpal) arthritis (capitate migrates proximally into the SL gap)**, **Stage IV: pancarpal (radiolunate rarely involved)**.
- **Radiolunate joint is spared** (spherical, congruent; load is perpendicular): **basis for salvage procedures that preserve it**.
- **Treatment: stage I: radial styloidectomy ± SL reconstruction/denervation**; **stage II: PRC or scaphoid excision + four-corner fusion (4CF)**; **stage III: 4CF** (PRC contraindicated: capitate head damaged); **stage IV: total wrist arthrodesis** (or arthroplasty in low-demand).
- **PRC vs 4CF: similar pain relief and grip**; **PRC: more ROM, simpler, no union needed, but more radiocapitate arthritis long-term**; **4CF: better grip/durability, longer immobilisation, non-union, dorsal impingement (lunate in DISI must be corrected)**.
- **Chondrocalcinosis (CPPD)** also causes SLAC-like changes (SLAC from CPPD; STT arthritis).

10. Lunotriquetral (LT) dissociation

- **FOOSH on the hypothenar region with wrist extension and radial deviation**; **ulnar-sided wrist pain**, **Reagan ballottement test**, **Kleinman shear test**, **Derby test**.
- **VISI (volar intercalated segment instability: lunate flexed, follows the scaphoid)** only if the dorsal radiocarpal and DIC ligaments are also injured; **break in Gilula arcs** (LT).

- **Differential: TFCC tear**, **ulnar impaction**, **ECU tendinopathy/subluxation**, **pisotriquetral arthritis**.
- **Treatment: splint, arthroscopic debridement, pinning**, **LT ligament repair/reconstruction (ECU tenodesis)**, **LT arthrodesis**, **ulnar shortening osteotomy** (tensions the ulnocarpal ligaments; especially with positive ulnar variance).

11. Midcarpal instability (CIND)

- **Palmar midcarpal instability (commonest CIND)**: **painful "clunk" with ulnar deviation and pronation** (proximal row snaps from flexion to extension); **generalised laxity**; **VISI at rest**. **Midcarpal shift test (Lichtman)**. **Treatment: splints (pisiform boost), proprioceptive training (FCU, ECU strengthening)**, **dorsal capsulodesis/ligament reconstruction**, **midcarpal (4CF) fusion** if refractory.

12. Perilunate dislocation and fracture-dislocation

- **High-energy hyperextension, ulnar deviation, intercarpal supination.**
- **Mayfield progressive perilunar instability (radial to ulnar around the lunate):**
  - **Stage I: scapholunate disruption** (or scaphoid fracture),
  - **Stage II: capitolunate dislocation** (through the **space of Poirier**),
  - **Stage III: lunotriquetral disruption** (**perilunate dislocation**: capitate dorsal to the lunate; lunate stays in the radius fossa),
  - **Stage IV: lunate dislocation** (**dorsal radiolunate ligament fails**: lunate **extruded volarly into the carpal tunnel** ("**spilled teacup sign**" on lateral), **acute median nerve compression**).
- **Greater arc injuries (through bones: trans-scaphoid, transcapitate (scaphocapitate syndrome: Fenton: capitate head**

rotates 180 degrees), trans-triquetral) vs **lesser arc (pure ligamentous)**; **trans-scaphoid perilunate fracture-dislocation (de Quervain) = commonest**.

- **PA radiograph: disrupted Gilula arcs**, **triangular (instead of trapezoidal) lunate ("piece of pie" sign)** in lunate dislocation, overlapping carpal bones.
- **Missed in about 25%** initially (**review the lateral: capitate-lunate-radius colinear?**).
- **Management: urgent closed reduction (Tavernier manoeuvre)** (**median nerve symptoms = emergency**); **definitive: open reduction (dorsal ± volar approach), repair of SL/LT ligaments, fixation of fractures (scaphoid screw), K-wires (SL, LT, scaphocapitate), carpal tunnel release (volar approach)**. **Outcomes:** post-traumatic arthritis (**about 50%+ radiographically**), stiffness, but acceptable function.
- **Chronic (over 6-8 weeks) missed perilunate dislocations:** open reduction possible (± shortening); **PRC** or **wrist arthrodesis** as salvage.

13. Kienböck disease (lunate AVN) (brief)

- **Young men (20-40)**, **negative ulnar variance (association)**, **single vessel supply (Y/I pattern)**, repetitive loading.
- **Lichtman staging: I normal radiograph (MRI changes)**, **II lunate sclerosis**, **IIIA collapse without carpal collapse (normal radioscaphoid angle)**, **IIIB collapse with fixed scaphoid flexion (ring sign; radioscaphoid angle over 60 degrees)**, **IIIC coronal fracture of the lunate**, **IV pancarpal arthritis**.
- **Treatment: stage I: immobilisation**; **II-IIIA: joint levelling (radial shortening osteotomy with negative ulnar variance; capitate shortening or radial wedge osteotomy with neutral/positive variance)**, **revascularisation (4,5-ECA pedicled vascularised graft)**; **IIIB: STT or scaphocapitate fusion, PRC**; **IV: wrist arthrodesis, PRC**

(if capitate head intact). **Children: non-operative (good prognosis)**.

- **Preiser disease** = idiopathic **scaphoid AVN**.

> Exam pearl: **Scaphoid: dorsal carpal branch of the radial artery enters at the dorsal ridge, retrograde to the proximal pole (AVN). Occult fracture: MRI. Displacement over 1 mm, intrascaphoid angle over 35, proximal pole = fix (headless screw, central axis; volar for waist, dorsal for proximal pole). SWIFFT: cast undisplaced waist fractures, fix non-unions.**

> Exam pearl: **Non-union with AVN (no punctate bleeding): vascularised graft (1,2-ICSRA; MFC free graft with humpback). SNAC: styloid-scaphoid, then scaphocapitate, then capitolunate; radiolunate spared. SLAC: same sparing. PRC needs intact capitate head; stage III = 4CF.**

> Exam pearl: **SL dissociation: Watson shift test, SL gap over 3 mm, ring sign, SL angle over 60-70 degrees, DISI; Geissler IV = drive-through. Mayfield I SL, II capitolunate (space of Poirier), III LT (perilunate), IV lunate dislocation (spilled teacup, median nerve). Greater arc through bones (trans-scaphoid commonest).**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.